Lexical Translation

Haus – house, building, home, household, shell

German Text — English Text

Step1: Collect statistics

Look at a parallel corpus (German Text vs English translation)

Haus	Count
house	8000
building	1600
home	200
household	150
shell	50

Step2: Estimate translation probabilities

Total = 8000 + 1600 + 200 + 150 + 50

= 10000

$$P_{Haus}(house) = \frac{8000}{10000} = 0.8$$

$$P_{Haus}(building) = \frac{1600}{10000} = 0.16$$

$$P_{Haus}(home) = \frac{200}{10000} = 0.02$$

$$P_{Haus}(household) = \frac{150}{10000} = 0.015$$

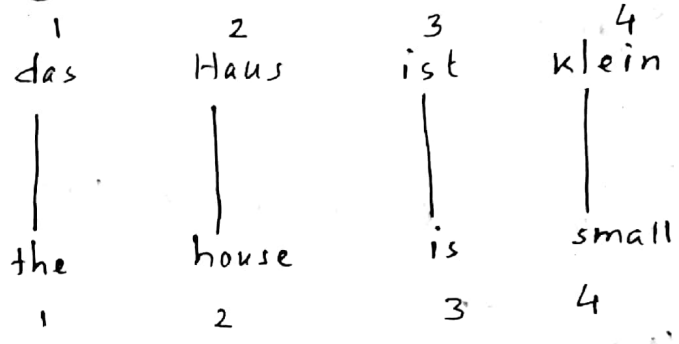
$$P_{Haus}(shell) = \frac{50}{10000} = 0.005$$

$$P_f(e) = \begin{cases} 0.8 & \text{if } e = \text{house} \\ 0.16 & \text{if } e = \text{building} \\ 0.02 & \text{if } e = \text{home} \\ 0.015 & \text{if } e = \text{household} \\ 0.005 & \text{if } e = \text{shell} \end{cases}$$

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Alignment:

In a parallel text (or when we translate), we align words in one language with the words in the other.



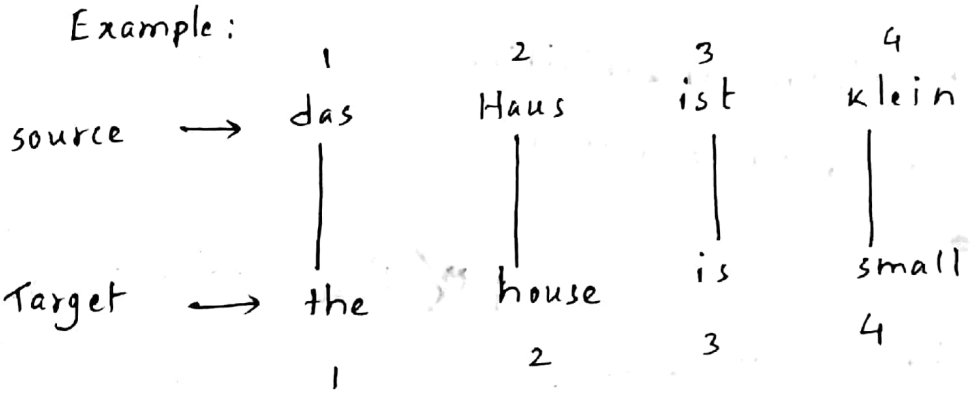
Word positions are numbered 1–4

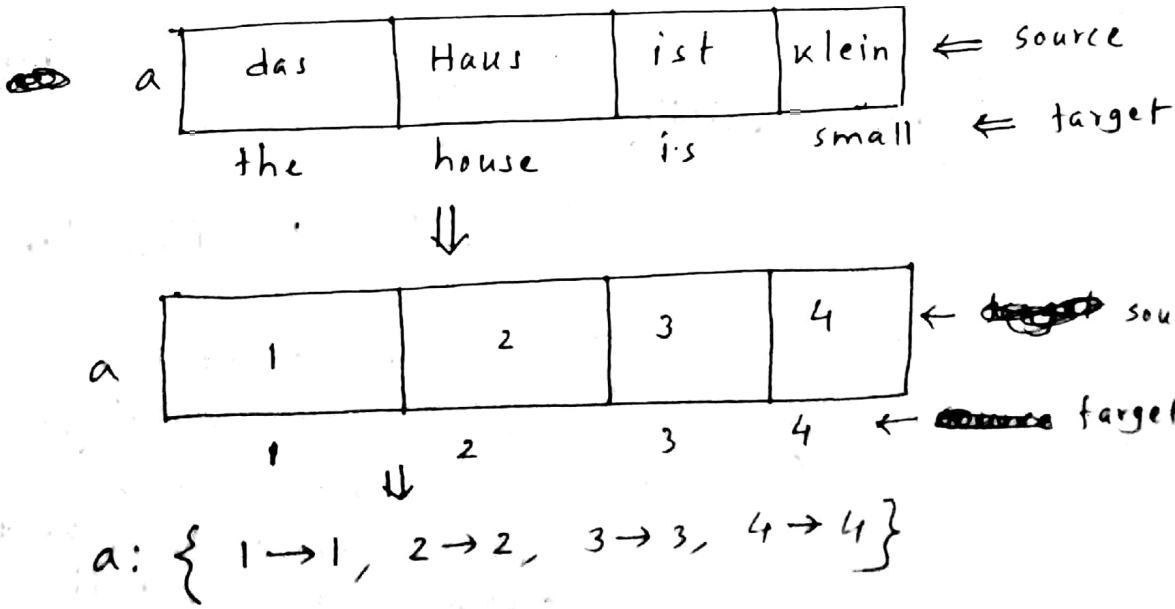
Alignment function

Mapping an target word (English) at position i to a source word (German) at position j with a function

$a: i \rightarrow j$

Example:



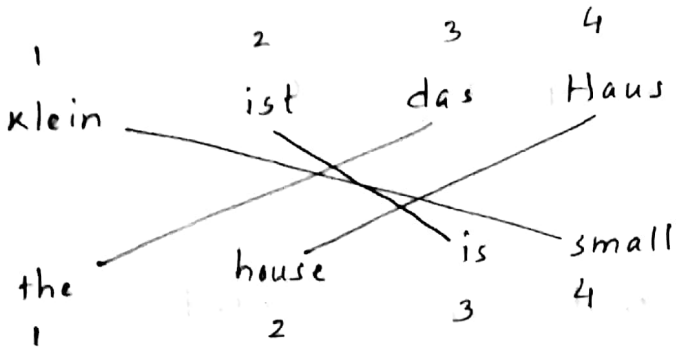


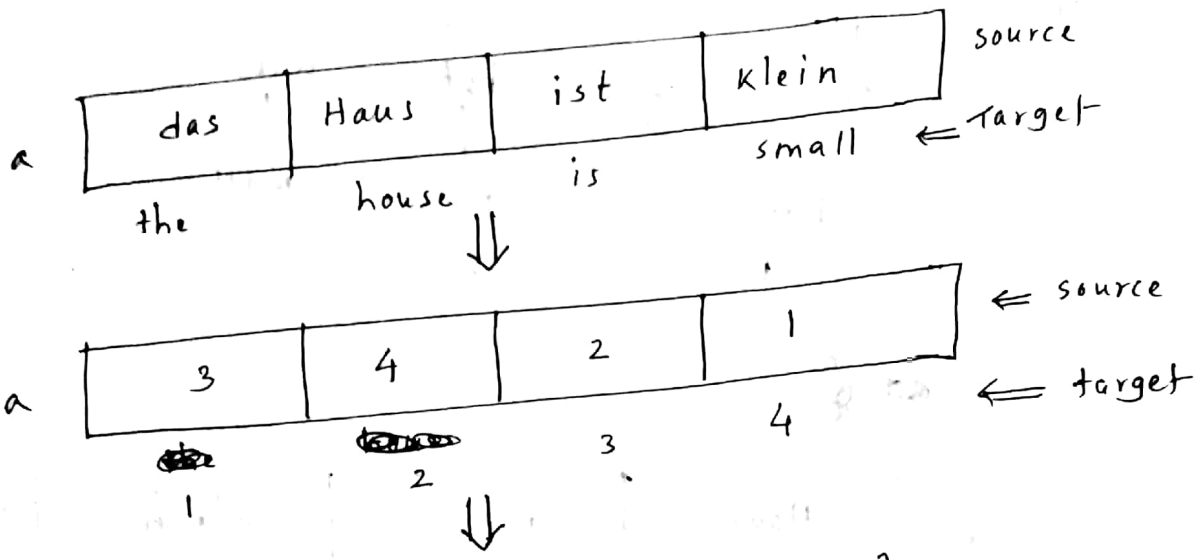
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ordering:

Words may be reordered during translation

Example:



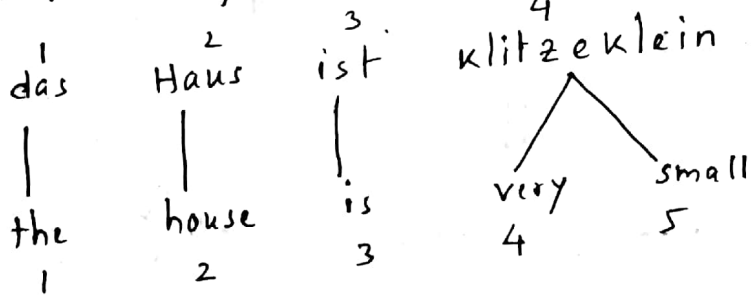


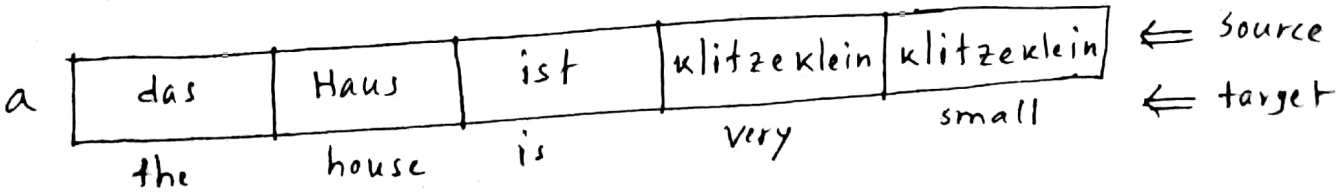
a: { 1→3, 2→4, 3→2, 4→1 }

One-to-many translation:

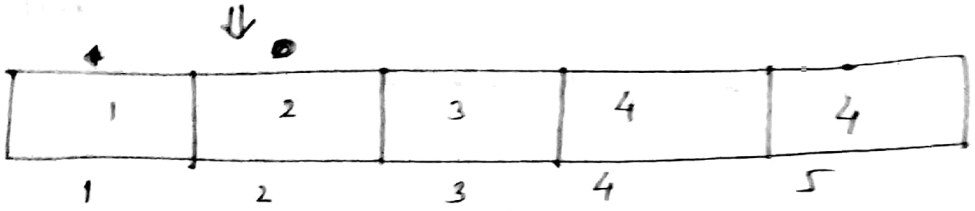
A source word may translate into multiple target words

Example:





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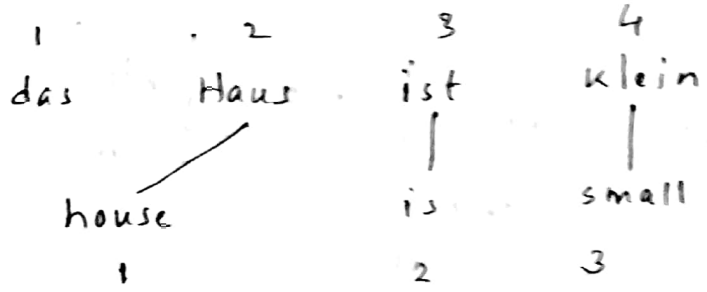


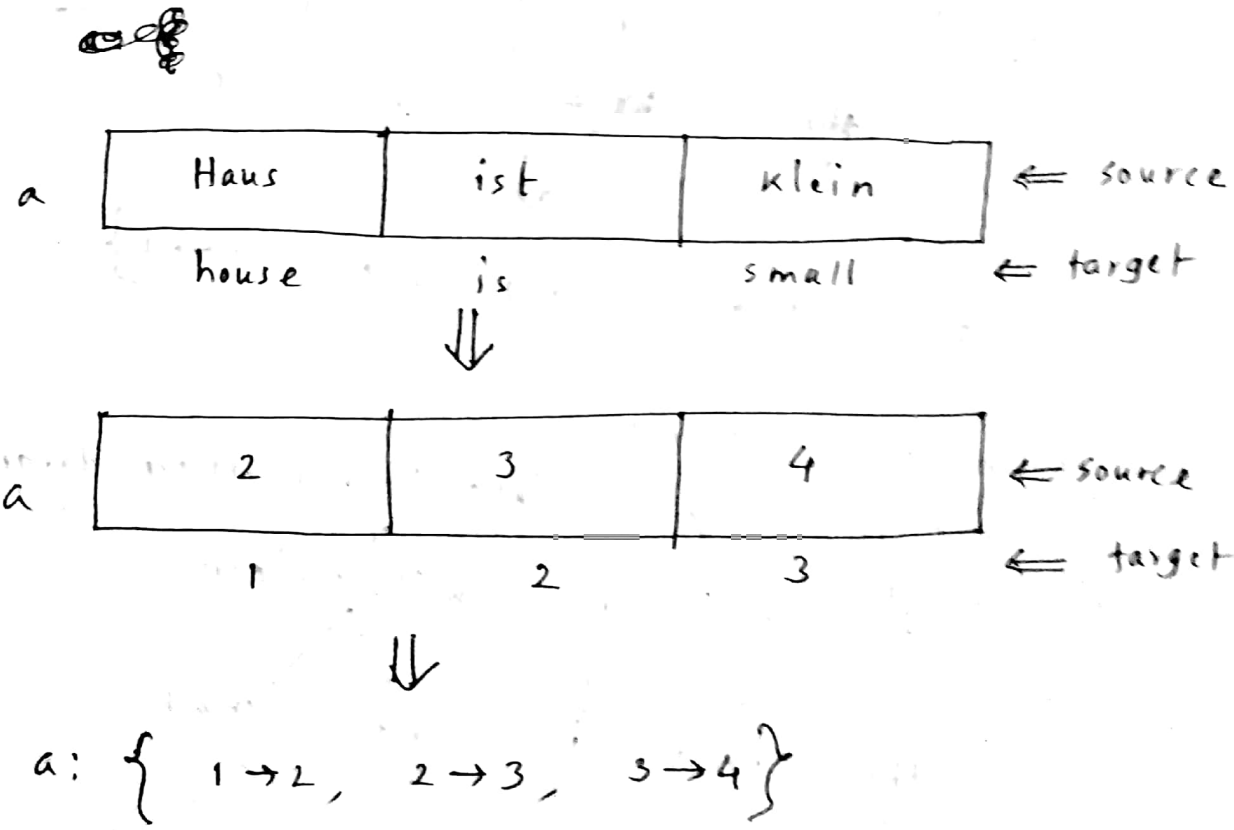
a : { 1→1, 2→2, 3→3, 4→4, 5→4 }

Dropping Words :

Words may be dropped when translated

Example :





a : { 1→2, 2→3, 3→4 }

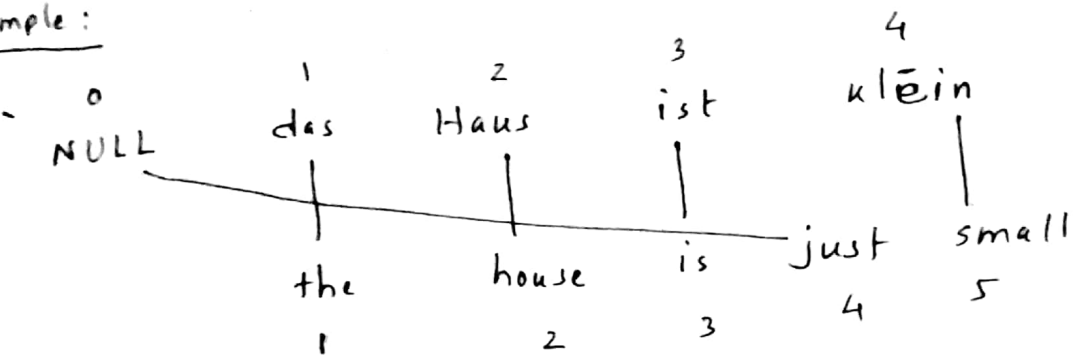
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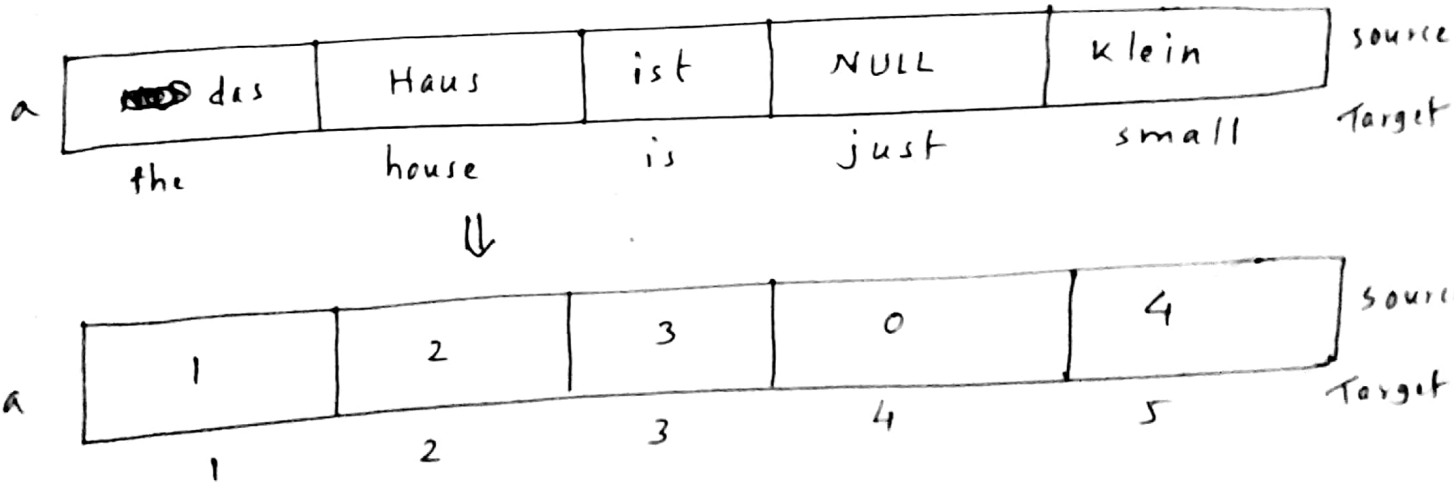
Inserting words:

Words may be added during translation

- The English just does not have an equivalent in German
- We still need to map it to something: special NULL token

Example:





$a: \{1 \to 1,\ 2 \to 2,\ 3 \to 3,\ 4 \to 0,\ 5 \to 4\}$

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